

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

NAME: A. Suraj Varma

Reg.No: 192525339

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Experiments (High)

Assessment Tool Weightage: 40%

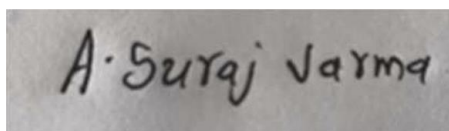
QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5

7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5
8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

*I **A. Suraj Varma (192525339)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q1. For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.

Answer: Given relation:

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Functional Dependencies:

1. $SID \rightarrow SName$
A student ID uniquely identifies the student name.
2. $CID \rightarrow CName$
A course ID uniquely identifies the course name.
3. $CID \rightarrow Instructor$
A course ID identifies the instructor of the course.
4. $(SID, CID) \rightarrow Grade$
A student's grade depends on both the student and the course.

Therefore:

$SID \rightarrow SName$
 $CID \rightarrow CName, Instructor$
 $SID, CID \rightarrow Grade$

Candidate Key:

Consider the attribute set (SID, CID).

$(SID, CID)^+$

$= SID, CID$
 $\rightarrow SName$
 $\rightarrow CName, Instructor$
 $\rightarrow Grade$

Therefore,

$(SID, CID)^+ = \{SID, SName, CID, CName, Instructor, Grade\}$

Since all attributes are obtained, (SID, CID) is a superkey.

Neither SID nor CID alone can determine all attributes.

Therefore, the candidate key is:

(SID, CID)

Conclusion: The functional dependencies are:

$SID \rightarrow SName$

$CID \rightarrow CName, Instructor$

$SID, CID \rightarrow Grade$

The candidate key is (SID, CID).

Q2. Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.

Answer:

First Normal Form (1NF) requires that every attribute contains only atomic values and that there are no repeating groups.

Unnormalized Relation:

STUDENT(SID, SName, Courses)

Example:

SID	SName	Courses
101	Rahul	DBMS, Java
102	Priya	Python, Networks

The Courses attribute contains multiple values in a single field. Therefore, the relation is not in 1NF.

Step 1: Identify the repeating group.

For student 101:

Courses = DBMS, Java

For student 102:

Courses = Python, Networks

Step 2: Separate the multiple values into individual rows.

1NF Relation:

SID	SName	Course
101	Rahul	DBMS
101	Rahul	Java
102	Priya	Python
102	Priya	Networks

Now every cell contains only one atomic value.

The combination (SID, Course) can be used as the primary key.

The relation is now in 1NF because:

1. All attributes contain atomic values.
2. There are no repeating groups.
3. Each row represents one student-course combination.

Conclusion:

The unnormalized relation is converted into 1NF by removing repeating and multi-valued attributes and storing each value separately.

Q3. Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.

Answer: Given:

R(A, B, C, D, E)

Functional Dependencies:

$A \rightarrow BC$

$BC \rightarrow D$

$D \rightarrow E$

Step 1: Find the candidate key.

Calculate A+:

$A^+ = \{A\}$

Using $A \rightarrow BC$:

$A^+ = \{A, B, C\}$

Using $BC \rightarrow D$:

$A^+ = \{A, B, C, D\}$

Using $D \rightarrow E$:

$A^+ = \{A, B, C, D, E\}$

Therefore:

$A^+ = \{A, B, C, D, E\}$

Since A determines all attributes, A is the candidate key.

Step 2: Check for partial dependency.

A is a single-attribute candidate key.

Partial dependency can occur only when a candidate key is composite.

Since the candidate key contains only A, there is no partial dependency.

Therefore, the relation is already in 2NF.

For a clear decomposition based on the functional dependencies:

$R_1(A, B, C)$

$R_2(B, C, D, E)$

Dependencies:

R_1 :

$A \rightarrow B, C$

R_2 :

$BC \rightarrow D$

$D \rightarrow E$

The second relation may require further decomposition for 3NF because $D \rightarrow E$ is a transitive dependency.

Conclusion:

The original relation is already in 2NF because its candidate key is the single attribute A and there is no partial dependency.

Q4. Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.

Answer: Given relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional Dependencies:

$\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

This is a transitive dependency because DeptName and ManagerID depend on DeptID, which depends on EmpID.

Step 1: Identify the problem.

DeptName and ManagerID should not be repeatedly stored with every employee.

Step 2: Decompose the relation into two relations.

EMPLOYEE:

EMPLOYEE(EmpID, EmpName, DeptID)

DEPARTMENT:

DEPARTMENT(DeptID, DeptName, ManagerID)

Step 3: Identify keys.

EMPLOYEE:

EmpID is the Primary Key.

DEPARTMENT:

DeptID is the Primary Key.

DeptID in EMPLOYEE is a Foreign Key referencing DEPARTMENT.

Final 3NF Schema:

EMPLOYEE

EmpID (PK)

EmpName

DeptID (FK)

DEPARTMENT

DeptID (PK)

DeptName

ManagerID

Conclusion:

The transitive dependency is removed by separating department information into a separate DEPARTMENT relation. The resulting relations satisfy 3NF and reduce redundancy and update anomalies.

Q5. Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.

Answer: Given:

R(A, B, C, D)

Functional Dependencies:

$AB \rightarrow C$

$C \rightarrow D$

$D \rightarrow A$

Step 1: Find candidate keys.

Calculate (AB)+:

$$(AB)^+ = \{A, B\}$$

Using $AB \rightarrow C$:

$$(AB)^+ = \{A, B, C\}$$

Using $C \rightarrow D$:

$$(AB)^+ = \{A, B, C, D\}$$

Therefore, AB is a candidate key.

Calculate $(BC)^+$:

$$(BC)^+ = \{B, C\}$$

Using $C \rightarrow D$:

$$(BC)^+ = \{B, C, D\}$$

Using $D \rightarrow A$:

$$(BC)^+ = \{A, B, C, D\}$$

Therefore, BC is a candidate key.

Calculate $(BD)^+$:

$$(BD)^+ = \{B, D\}$$

Using $D \rightarrow A$:

$$(BD)^+ = \{A, B, D\}$$

Using $AB \rightarrow C$:

$$(BD)^+ = \{A, B, C, D\}$$

Therefore, BD is also a candidate key.

Candidate Keys:

AB

BC

BD

Step 2: Check BCNF.

A relation is in BCNF if for every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey.

$AB \rightarrow C$:

AB is a candidate key, so this satisfies BCNF.

$C \rightarrow D$:

C is not a superkey, so this violates BCNF.

$D \rightarrow A$:

D is not a superkey, so this also violates BCNF.

Therefore, the relation is NOT in BCNF.

Step 3: Decompose using $C \rightarrow D$.

$R_1(C, D)$

$R_2(A, B, C)$

R_1 :

$C \rightarrow D$

C is the key of R_1 , so R_1 is in BCNF.

R_2 :

$AB \rightarrow C$

AB is the key of R_2 , so R_2 is in BCNF.

Final BCNF Decomposition:

$R_1(C, D)$

$R_2(A, B, C)$

Conclusion:

The original relation is not in BCNF because C and D are not superkeys for their respective dependencies. The decomposition into $R_1(C,D)$ and $R_2(A,B,C)$ produces BCNF relations.

Q6. Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.

Answer:

Consider:

$R(A, B, C)$

Functional Dependency:

$A \rightarrow B$

Decomposition:

$R_1(A, B)$

$R_2(A, C)$

Step 1: Create the Chase Table.

	A	B	C
R1	a	b	c1
R2	a	b2	c

The common attribute A has the same symbol a.

Step 2: Apply the functional dependency.

Given:

$A \rightarrow B$

Both rows have the same value of A.

Therefore, their B values must be equal.

$b = b2$

The table becomes:

	A	B	C
R1	a	b	c1
R2	a	b	c

Step 3: Check the result.

A row contains the original symbols:

a, b, c

Therefore, the decomposition is lossless.

Conclusion:

The decomposition:

$R \rightarrow R_1(A,B) \text{ and } R_2(A,C)$

is lossless because the Chase algorithm produces a row containing all the original symbols. Therefore, joining R1 and R2 reconstructs the original relation without producing spurious tuples.

Q7. Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.

Answer:

Given relation:

TEACHER(TID, Subject, Hobby)

A teacher can teach multiple subjects and can have multiple hobbies. These two properties are independent of each other.

Therefore, the multi-valued dependencies are:

$TID \twoheadrightarrow\twoheadrightarrow \text{Subject}$

$TID \twoheadrightarrow\twoheadrightarrow \text{Hobby}$

Example:

TID	Subject	Hobby
T01	DBMS	Cricket
T01	DBMS	Music
T01	Java	Cricket
T01	Java	Music

The relation contains unnecessary combinations between subjects and hobbies.

Step 1: Separate the independent multi-valued attributes.

Create:

TEACHER_SUBJECT(TID, Subject)

TEACHER_HOBBY(TID, Hobby)

TEACHER_SUBJECT:

TID	Subject
T01	DBMS
T01	Java

TEACHER_HOBBY:

TID	Hobby
T01	Cricket
T01	Music

Conclusion:

The multi-valued dependencies are separated into two relations. This removes unnecessary combinations and redundancy.

Final 4NF decomposition:

TEACHER_SUBJECT(TID, Subject)

TEACHER_HOBBY(TID, Hobby)

Q8. Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).

Answer:

Note: The question paper does not specify the actual relation for this question. The following is an illustrative Project-Supplier-Part example.

Consider the relation:

SUPPLY(Supplier, Part, Project)

Example:

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S1	P2	J2

The same supplier-part, supplier-project and part-project information may be repeated. This causes redundancy.

The relation can be decomposed using the following join dependencies.

1. SP(Supplier, Part)
2. SJ(Supplier, Project)
3. PJ(Part, Project)

SP:

Supplier	Part
S1	P1
S1	P2

SJ:

Supplier	Project
S1	J1
S1	J2

PJ:

Part	Project
P1	J1
P1	J2
P2	J1
P2	J2

The original relation can be reconstructed by joining the three decomposed relations when the join dependency is valid.

Conclusion:

Join dependencies can cause redundancy when several independent relationships are stored in one relation. Decomposing the relation into SP, SJ and PJ removes the redundancy and gives a 5NF design.

Q9. Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.

Answer:

Note: The question paper does not provide a specific poorly designed college schema. The following is an illustrative example.

Consider the relation:

```
COLLEGE(  
  StudentID,  
  StudentName,  
  DeptID,  
  DeptName,  
  CourseID,  
  CourseName,  
  Instructor  
)
```

Functional Dependencies:

$\text{StudentID} \rightarrow \text{StudentName}, \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}$

CourseID → CourseName, Instructor

1. Update Anomaly:

If the name of a department changes, it must be updated in multiple student records.

If some records are not updated, inconsistent department names may occur.

2. Insertion Anomaly:

A new department cannot be inserted unless a student is also entered.

Therefore, department information cannot be stored independently.

3. Deletion Anomaly:

If the last student of a department is deleted, the department information may also be lost.

Normalization to 3NF:

1. STUDENT

```
STUDENT(  
  StudentID PK,  
  StudentName,  
  DeptID FK  
)
```

2. DEPARTMENT

```
DEPARTMENT(  
  DeptID PK,  
  DeptName  
)
```

3. COURSE

```
COURSE(  
  CourseID PK,  
  CourseName,  
  Instructor  
)
```

4. ENROLLMENT

ENROLLMENT(
StudentID PK/FK,
CourseID PK/FK
)

Relationships:

DEPARTMENT 1 : M STUDENT

STUDENT M : N COURSE

The M:N relationship between STUDENT and COURSE is represented using ENROLLMENT.

Conclusion:

The original college relation contains insertion, deletion and update anomalies because different types of information are stored together. Decomposing it into STUDENT, DEPARTMENT, COURSE and ENROLLMENT removes redundancy and produces a better 3NF design.

Q10. Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.

Answer:

Note: The question paper does not provide the actual relation and functional dependencies for this question. The following is an illustrative example.

Consider:

R(A, B, C, D, E)

Functional Dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$D \rightarrow E$

Step 1: Calculate A^+ .

Start with:

$$A^+ = \{A\}$$

Using $A \rightarrow B$:

$$A^+ = \{A, B\}$$

Using $B \rightarrow C$:

$$A^+ = \{A, B, C\}$$

Using $C \rightarrow D$:

$$A^+ = \{A, B, C, D\}$$

Using $D \rightarrow E$:

$$A^+ = \{A, B, C, D, E\}$$

Therefore:

$$A^+ = \{A, B, C, D, E\}$$

Since A determines all attributes, A is a candidate key.

Step 2: Calculate B^+ .

$$B^+ = \{B\}$$

Using $B \rightarrow C$:

$$B^+ = \{B, C\}$$

Using $C \rightarrow D$:

$$B^+ = \{B, C, D\}$$

Using $D \rightarrow E$:

$$B^+ = \{B, C, D, E\}$$

A cannot be obtained.

Therefore, B is not a candidate key.

Step 3: Calculate C^+ .

$$C^+ = \{C\}$$

Using $C \rightarrow D$:

$$C^+ = \{C, D\}$$

Using $D \rightarrow E$:

$$C^+ = \{C, D, E\}$$

A and B cannot be obtained.

Therefore, C is not a candidate key.

Step 4: Calculate D^+ .

$$D^+ = \{D\}$$

Using $D \rightarrow E$:

$$D^+ = \{D, E\}$$

Therefore, D is not a candidate key.

Final Closure Results:

$$A^+ = \{A, B, C, D, E\}$$

$$B^+ = \{B, C, D, E\}$$

$$C^+ = \{C, D, E\}$$

$$D^+ = \{D, E\}$$

Candidate Key:

A

Therefore, the only candidate key for the relation is:

$\{A\}$

Conclusion:

The closure of A contains all attributes of the relation. Hence A is the candidate key. The other tested attribute sets do not determine all attributes and therefore are not candidate keys.